

AEGIS: Runtime-Guided GPU Task Collocator for Energy-Efficient AI Training

Ehsan Yousefzadeh-Asl-Miandoab⁽¹⁾ (ehyo@itu.dk), Buşra Demiray Karatay⁽²⁾, Pamela Delgado⁽²⁾, Pinar Tözün⁽¹⁾
(1) IT University of Copenhagen, (2) HES-SO University

1 GPU Underutilization: Causes and a Solution

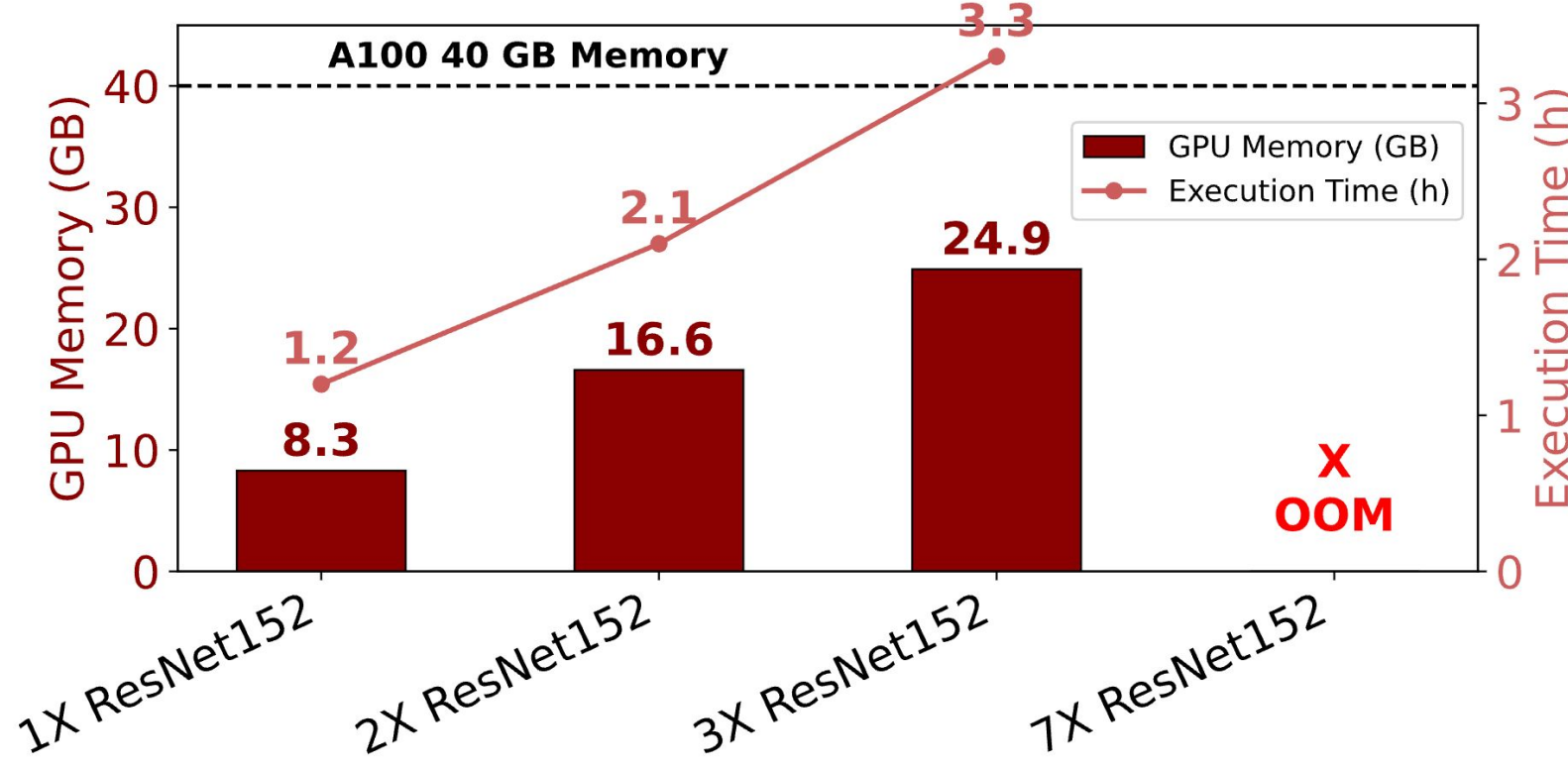
Real-world clusters exhibit only ~50% GPU utilization *

- 1- GPUs' lack CPUs-like **fine-grain sharing** and **virtual memory**
- 2- **Exclusive** GPU assignment by resource managers
- 3- **Black box** view of tasks and GPUs

Collocating tasks together increase GPU utilization!

* Yanjie Gao et al. "An Empirical Study on Low GPU Utilization of Deep Learning Jobs," ICSE'24.

2 OOM Crashes & Interference



GPU memory estimation can help for having robust collocation.

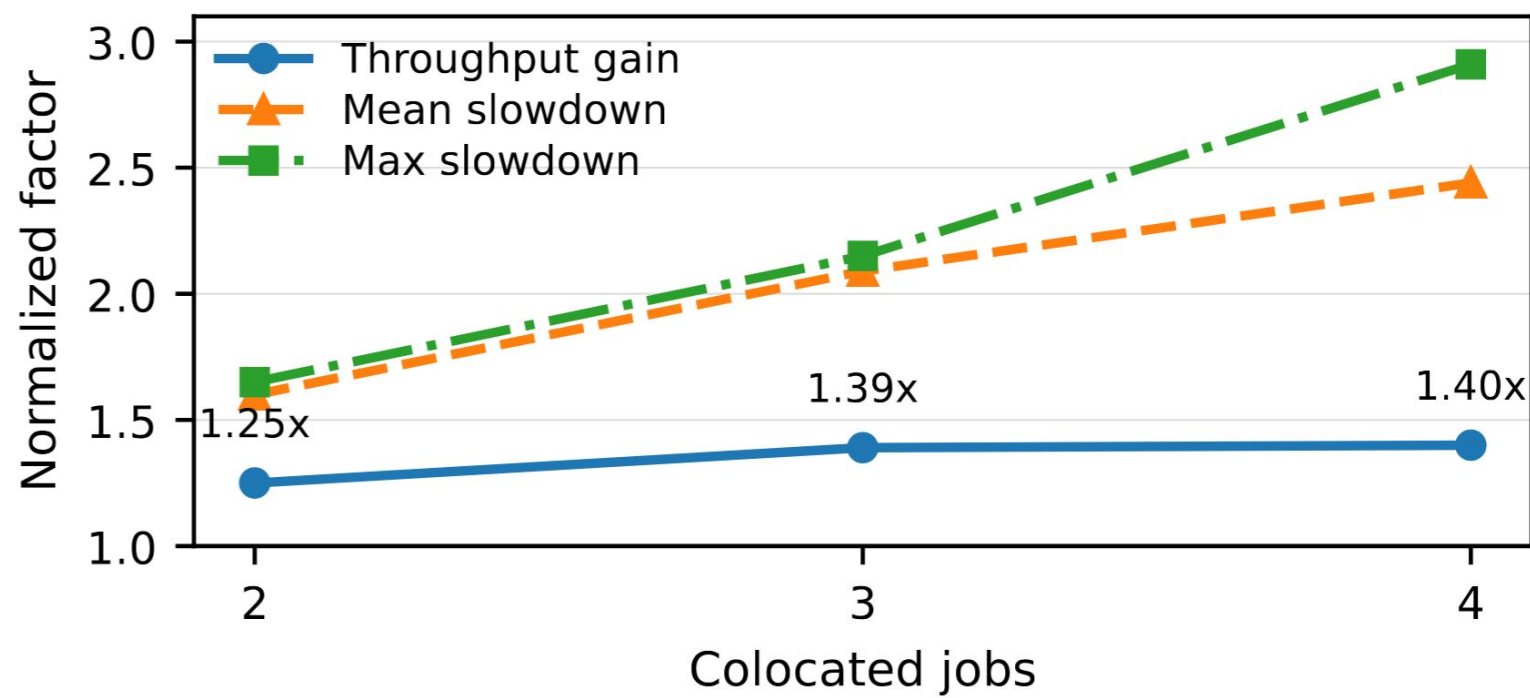
3 Estimating GPU memory: GPUMemNet

- Lightweight deep learning-based estimator.
- Synthetic, architecture-guided datasets.
- Rarely underestimates GPU memory need.



Machine Learning excels at pattern recognition.

4 Interference & Severe Slowdown

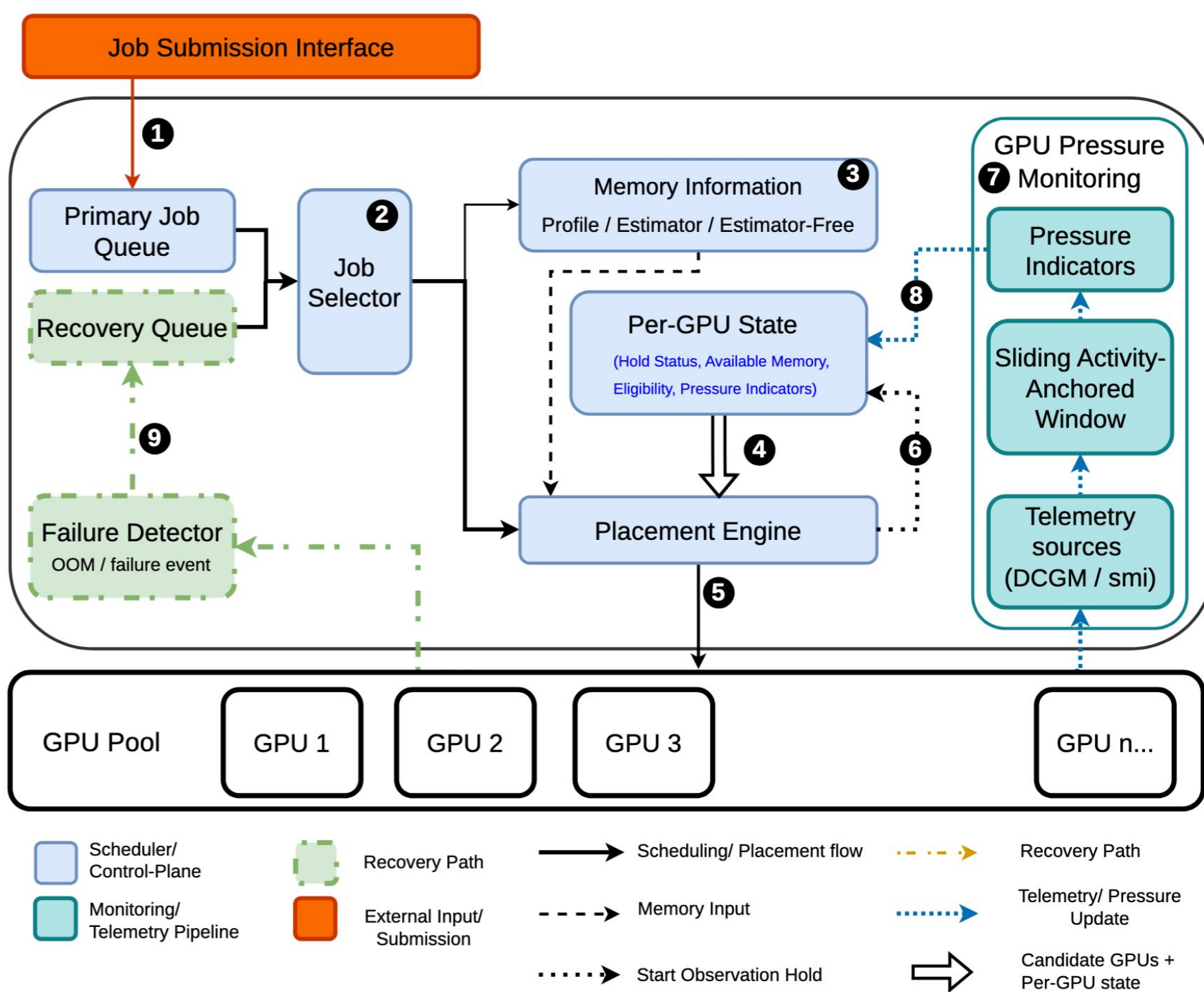


5 AEGIS Architecture

- Active monitoring & shortlisting saturated GPUs

$$R_{\text{SMACT}} \geq \tau_s \wedge (R_{\text{SMOCC}} \geq \tau_o \vee R_{\text{DRAM}} \geq \tau_d)$$

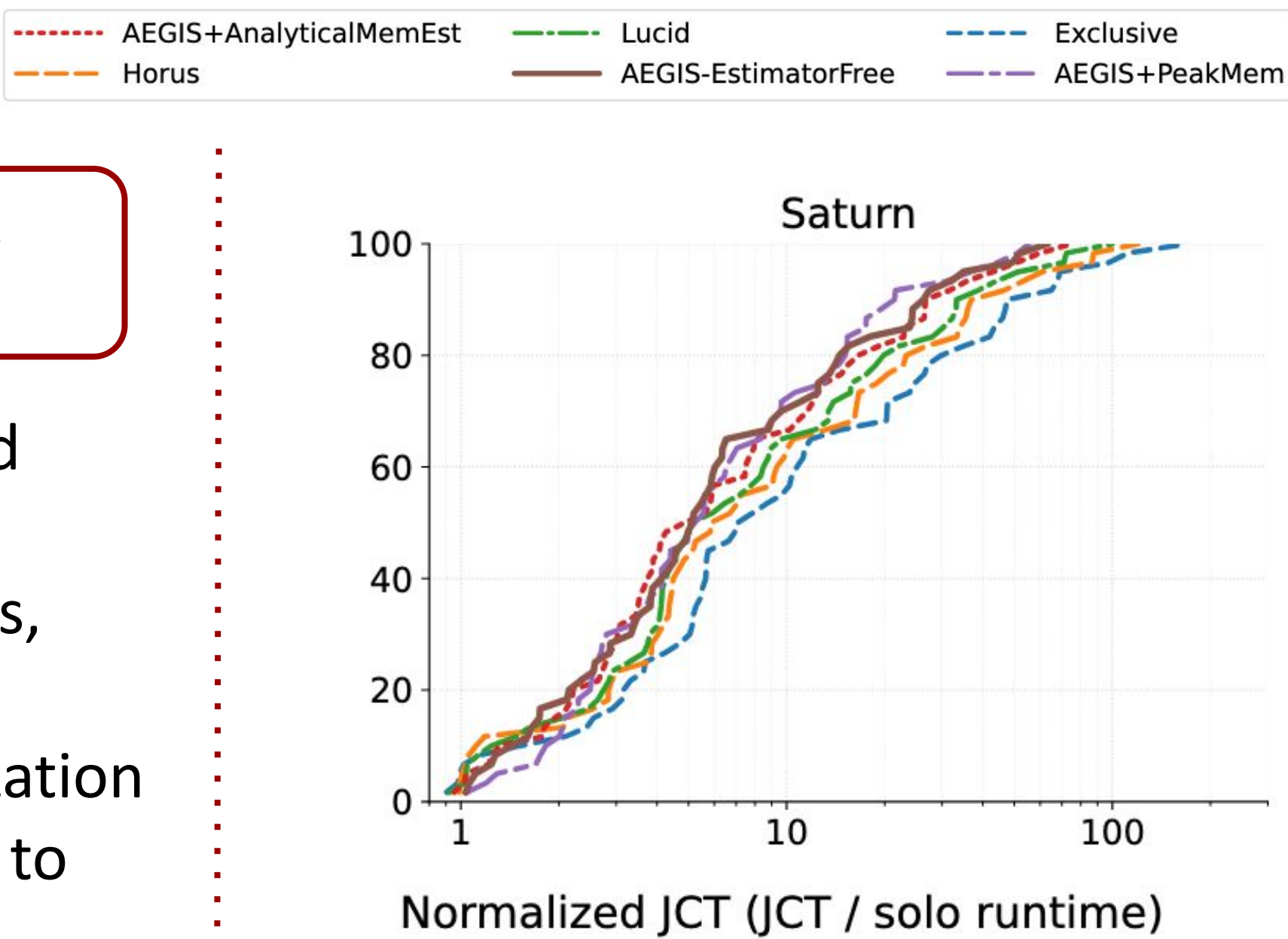
- Packing policies
 - Round Robin (RR)
 - Most Free -GPU- Memory (MFM)
- Recovery & relaunch when packing decision fails



6 Evaluation

- SW: 2X 60-task real-world traces of training tasks
- HW: A100 DGX Station with 3 GPUs

- Packing policies + preconditions + submitted workloads requirements define **#OOMs**.
- Tradeoff between profiling/ estimation costs, interference, and performance gains.
- AEGIS improves makespan (**27%**), GPU utilization (**23%**) and energy efficiency (**12%**) compare to exclusive case.



I am Ehsan.



Seeking academic & industry opportunities in computer systems and AI.